# Manufacturing Company Data Analysis Dashboard Project Report

### Project Overview

This project presents an interactive Manufacturing Company Data Analysis Dashboard developed using Python for data cleaning and analysis and Power BI for visualization. The dashboard analyzes machine performance, identifies failure trends, and provides insights to improve operational efficiency and reduce unexpected machine failures.

### Key Findings

- The dataset contains 10,000 machine records, with 339 machine failures identified.
- The average machine performance includes 39.99 Nm Torque, 1,540 RPM, 107.95 minutes Tool Wear, and 300 K Air Temperature.
- Heat Dissipation Failure (HDF) is the most common failure type, followed by Overstrain Failure (OSF) and Power Failure (PWF).
- The Torque vs RPM analysis shows an inverse relationship, where torque decreases as rotational speed increases.
- Product Type L has the highest production volume, indicating it is the most frequently manufactured product category.

### Recommendations

- Implement predictive maintenance based on machine sensor data to reduce downtime.
- Monitor tool wear, torque, and RPM regularly to detect abnormal machine behavior.
- Prioritize maintenance for machines prone to Heat Dissipation Failure.
- Use dashboard insights to optimize production planning and improve equipment reliability.

### Conclusion

The dashboard provides a clear view of manufacturing performance through interactive KPIs and visualizations. It enables quick identification of machine failures, operational trends, and production patterns, helping management make informed decisions to improve productivity, reduce maintenance costs, and enhance overall manufacturing efficiency.